NIHR | National Institute for Health and Care Research

NIHR Doctoral Award

Funding, Support or Development Opportunity	2025/445 NIHR Doctoral Award (Cohort 2)

SUMMARY INFORMATION

Application Title	Developing statistical methodology for coenrolment in platform designs
Contracting Organisation	University of Warwick
Start Date	01/10/2026
End Date	30/09/2029
Duration (months)	36
% FTE Commitment	100%
Percentage of award time undertaking practice/clinical work	0%
Total Research Costs Requested	£412,490.03
Has this application been previously submitted to this or any other funding body?	No
Partnership Fellowship	Not Applicable

NIHR Doctoral Award

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Mr Abraham Contreras
Organisation	Warwick Clinical Trials Unit
Department	Warwick Clinical Trials Unit
Professional Background	Not a Health or Care Practitioner
Detailed Professional Background	Statistician
Professional Registration	
ORCID	0000-0001-6449-4693
Research Degree Status	Not registered
Contextual Factors	not applicable

NIHR Doctoral Award

APPLICATION DETAILS

Scientific Abstract
Background
Coenrolment—the enrolment of participants in more than one interventional study—creates efficiencies in recruitment and trial delivery. It reduces competition between trials at the same research site competing for similar patient populations. This approach also enables simultaneous evaluation of multiple interventions throughout a treatment pathway. Coenrolment can improve evidence synthesis, accelerate policy translation, and offers to expand patient opportunities to participate in and benefit from clinical research. Despite its growing adoption in clinical trial research, there is a demand for updated, principled methodology.
Existing guidance predates contemporary trial designs, leaving unresolved challenges around interaction effects, bias, type I error control, and statistical power. Without robust frameworks, the benefits of coenrolment cannot be fully validated, limiting its credibility and implementation.
Aims and Objectives
The overarching aim is to develop statistical methodology for the design and analysis of clinical trials incorporating coenrolment. Specific objectives are i) to identify and characterise existing design and analytical approaches for coenrolment and analogous settings, ii) to review of current statistical practices of UK trial statisticians on trials permitting coenrolment, iii) to evaluate design and analysis methods by studying simulations of variations of coenrolment parameters, and iv) applied validation: to test and assess the developed methodology using real-world data from the CoReCCT platform, ensuring practical relevance.
Methods
WP1: Scoping Review – A systematic mapping of statistical methods relevant to coenrolment. This will produce a taxonomy of design types and analytical approaches.
WP2: Survey of Statisticians – A structured survey targeting UK trial statisticians to capture perspectives and current statistical practices in the design and analysis of coenrolment. Ethical approval will be obtained, and pilot testing will ensure clarity and validity.
WP3: Simulation Study – An investigation of coenrolment scenarios under varying design parameters (e.g., treatment combinations, interaction effects, outcome measure). I will explore both trial design and analysis models. Performance metrics will include bias, precision, type I error, and power.
WP4: Applied Analysis – Application of best-performing methods of WP3 to CoReCCT study data, generating estimates of treatment and interaction effects and comparing results with original analyses.
Dissemination and Impact
Findings from each work package will be disseminated through at least four open-access publications and presented at international conferences. Simulation code will be published and potentially developed into R packages for routine use. Methodological guidance derived from this work will inform NIHR-funded trials and shared across international research networks. Collaborative projects will be initiated and designed during a research placement abroad.
Research Inclusion
Patient and public involvement (PPI) is integral to this project. Previous focus groups and surveys indicate strong patient support for coenrolment, reinforcing its alignment with public preferences for broader treatment access and research participation. A dedicated advisory board will be regularly updated and consulted throughout the duration of the project and additional work will be undertaken to co-produce lay summaries with members of the public.

Plain English Summary
Background
Coenrolment is when a patient enrols and participates in more than one clinical trial study.
This gives patients more opportunities to take part in research and access new treatment options for

better health outcomes. Researchers also benefit from coenrolment as it helps trials improve their recruitment and research efficiency. Researchers can share their data with each other to study new treatment combinations which can lead to new insights into treatment plans and future research. Coenrolment is widely accepted among the scientific community and believe that coenrolment should be considered more routinely when designing a clinical trial. Despite this support, the available data on coenrolment is still limited and there is currently no clear statistical method to fully understand the results of clinical trials that have allowed for coenrolment. More research is also needed with members of the public, especially those involved in the consent process such as parents, family members, and other substitute decision-makers.

Research Aim

To develop statistical methods to analyse data from coenrolment trial designs and improve its understanding and accessibility across clinical trial research.

Project Plan

I will carry out four studies to achieve my research aim.

Study 1: I will review the research literature on statistical methods for analysing coenrolment in clinical trial research and understand all the key factors to consider when analysing coenrolment data. I will also look at other statistical models that can work for analysing patient data with more than one intervention.

Study 2: I will carry out a survey on UK trial statisticians to understand the current practice for handling coenrolment in clinical trial research. I will identify knowledge gaps and areas of concern when analysing coenrolment in clinical trials.

Study 3: I will carry out a simulation study to test different trial designs with coenrolment and their effect on the reliability of trial results. I will also test different ways of analysing trial data with coenrolment. I will use data from CoReCCT (the Confederation of Respiratory Critical Care Trials) – a group of 4 large critical care trials that allow for coenrolment across their studies. I will measure and compare the statistical performance of each trial design and analysis method.

Study 4: I will use data from the CoReCCT studies and apply the best statistical method identified in Study 3. I will compare these results with the original CoReCCT analyses.

Dissemination & Impact

I will share findings through research papers, presentations at academic conferences, international research placements, and public engagement events with the local community. The knowledge gained from developing this statistical methodology can directly impact the ongoing trials in CoReCCT and future clinical research around the world that allow for coenrolment.

Research Inclusion

An advisory board of lay members will be consulted twice per year and will undertake appropriate training to maximise the diversity and inclusion of all involved and affected by my research. I will deliver two workshops of 10-20 participants from the public to co-produce plain language descriptions of coenrolment and create a glossary of terms. This can be used by researchers to update their patient-facing material and help patients to understand coenrolment in order to provide informed consent, reducing burden and improving accessibility.

Working with People and Communities Summary

I will form a Patient and Public Involvement (PPI) Advisory Board of 2-4 participants. I will extend my search to wider public engagement networks within my university and through social media. There will be no restrictions or requirement of a lived experience for this advisory board as the research has the potential to affect patients, their family members, and substitute decision-makers. We will consult the PPI Advisory Board in the development of the planned workshops.

Co-production of a glossary of terms of key concepts in coenrolment trial designs

A preliminary PPI focus group was consulted to explore various ways of approaching this particular project. They recommended that a group of 10-20 participants should be consulted through in-person and online meetings to co-produce the glossary of terms. They believed this was a sufficient number to cover the diversity required to represent members of the public and characteristics that may affect their comprehension of trial terminology. In-person activities are included to maximise accessibility and inclusion of participants that may struggle with digital technologies. I will attend workshops by the PPI-SMART group from the NIHR Leicester Biomedical Research Centre as well as consult them on their

experience in developing lay glossary of statistical terms.
Target Population
Any member of the public will be eligible to participate. Strategies will be employed to optimise inclusion and diversity of characteristics such as age, socioeconomic status, and highest educational attainment. Existing evidence suggests that parents and family members (as well as those often approached for the consent) of patients are particularly underrepresented in this field of research.
Recruitment Strategies
Members of the public will be invited to participate in this research by sharing posters across notice boards (NIHR People in Research), specialised patient communities (ICUsteps, Independent Cancer Patients' Voice), social media, and established networks of public contributors at WCTU. Adverts will be posted for a maximum of 6 weeks. Public contributors will be randomly selected from the participants that have expressed interest.
Co-production Methods
I will conduct two workshops using the nominal group technique to produce definitions. Using the results of the scoping review and key factors highlighted by the survey of statisticians, I will produce a list of key terms to be defined in the workshops. For each key concept or terminology, the nominal group technique will facilitate consensus building towards a suitable plan language description of the term. The first stage involves sending the questions in advance and letting participants reflect and record their proposed definitions. The second stage will allow participants the opportunity to present their plain language descriptions, as well as ask for clarifications or refine/group their ideas together. In the third stage, the participants will be asked to rank the presented plain language descriptions, and the score will aid in determining a consensus on which plain language descriptions are best.
Further Public Engagement
The preliminary PPI focus group was supported by the Warwick Institute of Engagement (WIE) Events and Activities Fund. This collaboration will be continued throughout the doctoral award and we will work together to prepare public engagement events to share research output to the local communities of the University of Warwick.

SUPPORT AND MENTORSHIP

Full Name	Dr Chen Ji
Organisation	Warwick Clinical Trials Unit
Email Address	C.Ji.3@warwick.ac.uk
Role	Mentor
Description of Support Provided	I will provide statistical support in the methodological development and its application in the related clinical trials. I will also support to Abraham's professional and personal development.

Full Name	Professor Nick Parsons
Organisation	University of Warwick
Email Address	nick.parsons@warwick.ac.uk
Role	Primary Academic Supervisor
Description of Support Provided	Primary supervisor of the PhD. Focussing mainly on statistical and methodological support. High level training in statistical software development and coding. For instance, the building of simulation models and implementation in R. The development of research skills, such as critical appraisal and report and paper writing.

NIHR Doctoral Award

Full Name	Professor Louise Rose
Organisation	King's College London
Email Address	louise.rose@kcl.ac.uk
Role	Academic Supervisor
Description of Support Provided	As an academic supervisor from a clinical academic background I will support Abraham in his patient and public facing activities as well as project conduct overall and dissemination and impact activities.

Full Name	Professor Ranjit Lall
Organisation	Warwick Clinical Trials Unit
Email Address	r.lall@warwick.ac.uk
Role	Academic Supervisor
Description of Support Provided	With my extensive experience and knowledge of complex intervention clinical trials and as the lead statistician for the CoReCCT study, I will provide statistical support for Abraham as an academic supervisor. I will also oversee his personal and career development during and after the fellowship.

SUMMARY BUDGET

Justification of Costs

Lead applicant salary at FA6 (points 29 to 33) from (https://warwick.ac.uk/services/peopleteam/internal/payroll/salscalescurrent/current/)
Travel and subsistence include travel costs to conferences and training courses. Costs are based on subsistence allowance policies at University of Warwick: (https://warwick.ac.uk/services/peopleteam/internal/payroll/expenses/uk1/subsistence/)
Conference 1: ANZICS-CTG (Noosa) fee AU$1815, accommodation for 3 nights AU$1050, fees based on 2026 costs (https://www.anzics.org/wp-content/uploads/2025/06/Noosa-Brochure-2026.pdf)
Conference 2: ICTMC 2028, registration fee £475 based on 2026 costs (https://ictmc.org/registration-2/)
Training Courses – fees
Practical Survey Design and Web Based Methods, University of Surrey - £495 (https://www.surrey.ac.uk/cpd-and-short-courses/practical-survey-design-and-web-based-methods#fees)
Workshop on Adaptive and Bayesian designs in real trials: clinicians' patients' and statisticians' perspectives (2-day in-person workshop), MRC Biostatistics Unit - £40
Using Simulation Studies to Evaluate Statistical Methods (4-day online training), UCL ICTM - £372 (https://www.ucl.ac.uk/population-health-sciences/clinical-trials-and-methodology/study/short-courses/using-simulation-studies-evaluate-statistical-methods)
Patient and Public Involvement short course, UCL ICTM - £323 (https://www.ucl.ac.uk/population-health-sciences/clinical-trials-and-methodology/study/short-courses/patient-and-public-involvement)
Effective Community Engagement, The Centre for Ethnic Health Research - £400 (https://ethnichealthresearch.org.uk/training/effective-community-engagement-2/)
Focus group, 10 people, 2 session, 1.5 hours (plus 1 hour prep) per session, £27.5 per person - £687.5 per session total cost (second session includes travel costs for participants)
PPI Advisory board, max 4 members, 2 hours of activity, twice per year - £880 total cost per year
Research placement accommodation costs based on Monash University (ANZIC-RC) visiting researcher (https://www.monash.edu/accommodation/staff-and-visitor/staff-and-visitor-pricing)
PhD Research Studentship Fees (https://warwick.ac.uk/services/finance/studentfinance/fees/pgr/)

NIHR Doctoral Award

ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	not applicable

Lead Applicant / Co-applicant Name:

Abraham Contreras

Module 1 – Contributions to the generation of new ideas, tools, methodologies or knowledge

My first research role was during my undergraduate studies where I completed a 5-month part-time internship at the Alan Turing Institute contributing towards the modelling of an AI-based diagnosis of early-onset dementia. I conducted a literature review on human spatial navigation, trajectory data and spatio-temporal forecasting studies, and presented at weekly lab sessions. I used my skills in statistical programming to create data processing and analysis workflows, cleaning large datasets of over 3 million rows and implementing artificial neural networks to classify socio-demographic characteristics of individuals based on trajectory data. I was exposed to key aspects of research, learning the importance of defining a research question and aim, in addition to technical skills such as exploratory analysis and data visualisation. Being able to use my statistical skills to contribute towards health research led me to seek roles in medical statistics where in July 2021, I was appointed as a Research Associate at the Warwick Clinical Trials Unit.

In 2022, I received the NIHR Predoctoral Fellowship Award where I completed an MSc in Statistics for Clinical Trials and undertook rotations across the Warwick Clinical Trials Unit, shadowing trial managers, health economists, programmers, and statisticians. This exposure gave me a comprehensive understanding of the many facets of clinical research as well as equipping me with the necessary skills to become a versatile and capable statistician in clinical research. I have since been able to contribute meaningfully to the EPPiC (Emergency, Prehospital, Perioperative and Critical Care) research group, providing statistical input to the protocols and statistical analysis plans for POSED (doi: 10.1016/j.resplu.2022.100310) and AIRWAYS-3. I have also supported the development of novel statistical models for the OHCAO (Out-of-Hospital Cardiac Arrest) Registry Database as well as the i-WOTCH study. I have gained experience across the full lifecycle of a clinical trial—from its initial design and setup through to its delivery, analysis, and publication.

I continued to collaborate with the OHCAO Registry Database team throughout my fellowship award. This led to the development of a full research study where I identified and consulted the appropriate experts, coordinated contributions, and led a multidisciplinary team to conduct a novel application of time series methodology to out-of-hospital cardiac arrest (OHCA) observational data. I was also responsible for the formal analysis, methodology, investigation, writing the original draft, reviewing and editing. This work resulted in a first-author paper publication (doi: 10.1016/j.resuscitation.2025.110823) in Resuscitation; impact factor of 6.5 and number one journal ranking in emergency medicine in 2024. The novel approach provided new insights into the epidemiology of OHCA outcomes during pandemics, offering evidence that may inform how to strategically and effectively implement future policies to minimise short- and medium-term impacts during public health emergencies. This demonstrated my ability to lead research from concept to publication.

Since completing the predoctoral fellowship, I have been a trial statistician for the Confederation of Respiratory Critical Care Trials (CoReCCT) comprising four large critical care trials along a single treatment pathway and incorporates a novel platform-style approach to coenrolment across the domains of CoReCCT. I conducted extensive literature

reviews on coenrolment methodology, consulted a range of experts in data science and trialists in specialties such as cancer, surgery, and emergency care. I also delivered a workshop at the NIHR Stats Group annual conference to gauge current statistical practices on coenrolment as well as to present the study design of CoReCCT. Since December 2025, all four trials have been open for recruitment. A paper providing an overview of CoReCCT was published (doi: 10.1016/j.bja.2024.10.033) in the British Journal of Anaesthesia (BJA); impact factor 9.2 and number one journal ranking in anaesthesiology in 2024. I contributed a summary of statistical and methodological challenges faces in CoReCCT. I also provide statistical advice for the domain protocols and statistical analysis plans (SAPs).

Module 2 – Contributions to the development of others and maintenance of effective working relationships

At the Warwick Clinical Trials Unit, I am responsible for mentoring newly joined statisticians and team members, to help familiarise themselves with the unit and colleagues. I also peer mentor for the NIHR RSS pilot mentoring programme where I create monthly topics for discussion to share experiences and advice regarding personal and career development.

Despite being a research focused staff, I ensure that I regularly contribute towards the excellent teaching activities at the Warwick Medical School such as delivery of statistics lectures, workshops, exam invigilation, and marking of assessments. I have continual responsibilities with the OHCAO team with plans to further develop my MSc dissertation into a paper. I continue to provide statistical advice to grant applications, annual epidemiology reports, survey analysis, and data visualisation and presentation. I also contribute to standard operating procedures (SOPs), statistical processes, and the business continuity plan, collecting and raising the views of my peers to senior leadership.

I have established a collaboration agreement with the Australian and New Zealand Intensive Care Research Centre (ANZIC-RC) and have planned to undertake a 12-week research placement. In consulting experts in trial methodology, I have also secured interest in a future collaborative project with the lead methodologist, Dr Anna Heath, of the PRACTICAL platform study. This work is expected to build upon the output of my doctoral research to develop and validate statistical analysis methods for platform trials.

During my postgraduate studies, I assisted in providing feedback of the modules as the first cohort, raising the concerns of my peers as a course representative and providing a voice for them. I also produced and contributed to their promotional activity and continue to nurture this working relationship through recommendations, invitations to presentations, and occasional updates of my work.

Module 3 – Contributions to the wider research community

Despite my early stage in my research career, I have been invited to peer review for Resuscitation Plus, providing my expertise in time series modelling of cardiac arrest data. In 2024, I was invited to speak at the International Society of Clinical Biostatistics (ISCB) in Thessaloniki, Greece, where I provided a summary of my novel research into coenrolment methodology.

I am also a member and regular participant of the Warwick Institute of Engagement (WIE) Network, the Royal Statistical Society West Midlands local group, and the Warwick R

(programming language) User Group. I am part of a working group aiming to develop a local Patient and Public Involvement and Engagement (PPIE) resource hub dedicated to the Warwick Medical School.

I regularly present at the Clinical Sciences Research Laboratory (CSRL) Academic Seminars based at the University Hospital of Coventry and Warwickshire (UHCW. I disseminate research findings regarding coenrolment but also provide summaries of trial methodology to improve accessibility of statistical methods and benefit a wide range of participants, from clinicians to PhD students. I also provide feedback to other researchers presenting their proposals and projects.

Module 4 – Contributions to broader society

I carry a personal ambition to improve mathematical literacy and the understanding of statistics of the general public. I have provided Maths and English tutoring for GCSE students with Team Up Hub, restoring the learning gap incurred as a consequence of the pandemic lockdowns. I taught over 16 pupils throughout 5 programmes and contributed towards an average of +1 grade of progress per programme. I worked with Coram children's charity to provide A level Maths tutoring to students with refugee, asylum seeker, or undocumented status, improving access to higher education for those in less fortunate positions. I now volunteer at a local afterschool club, teaching Maths at key stage 2. My extensive volunteering experience has dramatically improved my communication skills and have acquired the ability to translate complex concepts to be understood by a wide variety of audiences.

I work closely with the WIE to develop and carry out public engagement work. We have since conducted a focus group pivotal in developing the research proposal for this application and plan to share the outputs of future research to local communities through evening talks and public stalls.

I am recognised and known as a medical researcher at my local church community and actively help in translating research news into lay summaries. This was particularly helpful in easing concerns regarding the trial results and release of vaccines.

Module 5 - Your suitability for this award

I am committed to becoming an internationally recognised expert in the statistics and trial methodology of coenrolment, and this award will provide the essential platform to establish myself as an independent researcher. My long-term goal is to develop a principled statistical framework for the design and analysis of multiple intervention effects per patient, enabling clearer interpretation and accessibility, greater efficiency, and maximising scientific value of clinical research.

My experience to date demonstrates both my capability and readiness for doctoral research. I have contributed to methodological innovation across several major trials and registries, led a full research study to publication as first author, and provided statistical expertise within CoReCCT, where coenrolment is central to its design. My record of mentoring, teaching, peer review, and public engagement further demonstrates the breadth and maturity of my research profile. This award will allow me to build the methodological foundations necessary to advance coenrolment methodology and position myself as a leader in this emerging field.

Background and rationale

Clinical trials are increasingly complex, resource intensive and often conducted concurrently within the same research site. As a result, multiple studies frequently compete for similar patient populations, creating challenges for recruitment and the efficient delivery of research(1). Improving the efficiency and productivity of trials is therefore a key priority for NIHR and the wider methodological community(2).

Coenrolment—the enrolment of a participant in more than one interventional study simultaneously or sequentially—offers an important opportunity to enhance efficiency(3). By enabling the assessment of multiple interventions within the same participant, coenrolment facilitates more comprehensive evaluations of treatment combinations and pathways than traditional single-trial designs(4). This can be operationalised through platform trials, the use of master protocols, or explicit coenrolment agreements between independent trial teams.

For patients, this coenrolment can increase access to potentially beneficial treatments, while for researchers and policymakers it provides richer evidence to support timely and informed decision-making(5). Despite these advantages, significant methodological and practical uncertainties remain.

Key Problem 1: Lack of modern statistical methodology for coenrolment

The only published statistical guidance for analysing data from co-enrolled participants dates back to 1996 (6), predating the emergence of contemporary platform trial designs and master protocols. A number of unresolved methodological challenges remain, including how to address interaction effects, bias, type I error rate control, statistical power, and appropriate modelling structures(7). These issues are rarely articulated in current statistical analysis plans for trials permitting coenrolment(8), highlighting the urgent need for updated, principled methodology.

Key Problem 2: Inconsistent reporting and implementation of coenrolment

Although there is broad support within the research community for the use of coenrolment, current reporting practices are highly variable and often incomplete(9). Important information—such as the criteria governing eligibility, frequency and timing of coenrolment, and handling of treatment interactions—is frequently absent from trial protocols and final reports. This lack of transparency limits the interpretability and generalisability of findings and constrains rigorous evaluation of coenrolment as an efficiency-enhancing strategy(10).

Research Importance

The absence of robust, up-to-date methodology and guidance hinders the advancement of coenrolment as a credible and efficient trial methodology. Without a clearer statistical framework, it is difficult to validate its claimed benefits—such as improved evidence synthesis, long-term efficiency gains, and accelerated policy translation.

To inform the development of this application, I conducted a focus group with eight members of the public, including former patients who had been enrolled in multiple trials. Participants emphasised that coenrolment expands “access to treatments”, particularly for individuals with limited therapeutic options, and expressed strong support for research aimed at strengthening coenrolment methodology. They highlighted the need to “change the narrative” around coenrolment so that it is recognised as something routine, rather than as something unusual or potentially problematic.

Patient perspectives also support the acceptability of coenrolment. Surveys of individuals affected by breast cancer and parents of premature infants show that a majority–64% and 74% respectively–would consider enrolling themselves or their child into more than one trial(11, 12). These findings suggest that coenrolment aligns with patient preferences for broader access to potential treatments and opportunities to contribute to research.

Evidence Review

Coenrolment has been well studied in critical care and HIV research(13), but its application is now expanding across a wider range of clinical specialties. For example, in orthopaedics, researchers have proposed the WHiTE coenrolment platform concept (14) which enables multiple simultaneous comparisons. This approach provides the opportunity to generate high-quality evidence for multiple interventions and across stages of care with one study instead of several trials.

In oncology, the Add-Aspirin trial team demonstrated the feasibility and value of coenrolment in maximising recruitment efficiency as well as improving opportunities for patients to participate in and benefit from clinical research(15). By establishing coenrolment agreements with trials of primary therapies, the researchers were able to assess an intervention common across the treatment pathways of different tumour types. They found coenrolment to be safe, scientifically valid, and ethically acceptable when planned properly, prompting calls for more routine consideration of coenrolment in cancer trials.

Coenrolment is also increasingly incorporated into platform designs, such as the PRACTICAL platform study(4) which also enables multiple comparisons of interventions across multiple therapeutic domains. Each “domain” functions as its own trial, studying its own set of interventions that target the same disease mechanism but exists under the overarching platform trial and its response-adaptive randomisation mechanism.

Another novel and versatile implementation of coenrolment is CoReCCT—the Confederation of Respiratory Critical Care Trials(16)—a UK research confederation following a platform-style approach.

I have been the trial statistician for CoReCCT since its inception in 2025.

This confederation comprises four coordinated domain trials across the care pathway of patients with acute hypoxemic respiratory failure (AHRF). Instead of a single platform study, the individual trials remain independent acting as “domains”

but work under a shared framework, enabling harmonised trial operations, shared data structures, and the evaluation of multiple interventions throughout a patient's treatment journey(17). Such designs illustrate how coenrolment can facilitate whole-pathway evidence generation in a way that is not achievable through independent conventional trials.

Despite its growing use, researchers consistently highlight the need for updated guidance on the statistical methodology and operational aspects of coenrolment. These include understanding how coenrolment policies influence selection bias, trial efficiency and costs(3), and generating empirical evidence on the statistical performance of coenrolment under realistic scenarios(10). There are also strong calls for greater engagement with stakeholders—including patients, statisticians, ethics committees, and family members involved in consent processes—to ensure that coenrolment is implemented transparently and ethically(18).

****Aims and objectives****

The overall aim of this fellowship is to understand the current uses and application of statistical and trial methods and to develop robust, modern statistical methodology and guidance to support the design and analysis of clinical trials that incorporate coenrolment. By advancing methodological frameworks and practical guidance, this work will improve the implementation, interpretation and transparency of coenrolment across clinical trial research.

The specific objectives are:

1. Evidence mapping—to identify and characterise existing design and analytical approaches used in coenrolment including statistical methods for analogous settings of multiple intervention effects.
2. Describe and summarise current statistical practices in clinical trial design and analytical approach where coenrolment is permitted, highlighting strengths and limitations
3. Methodological development—use simulation studies to evaluate variations in coenrolment trial designs, determining optimal approaches with respect to statistical power, type I error, interaction effects, and estimate precision
4. Applied validation—test and assess the statistical viability of the developed and recommended methodology using real-world data from the CoReCCT(16) domain trials, ensuring relevance and applicability to ongoing NIHR-funded research.

****Methodology / Plan****

Work Package 1—Scoping review of statistical methodology for planned coenrolment in randomised clinical trials

This work package will deliver a comprehensive scoping review of statistical methodology relevant to the design and analysis of clinical trials permitting planned coenrolment. The review will be conducted in accordance with JBI guidance for scoping reviews(19) and the protocol will be registered with the Open Science Framework (OSF) to ensure transparency and reproducibility(20). I will work closely

with specialist librarians at the University of Warwick to refine and validate the search strategy.

Scoping review questions:

- What assumptions and methodological factors are required for the design of experimental trials with multiple intervention effects per participant?
- What statistical methods have previously been applied in the analysis of trials permitting coenrolment, including analogous settings such as multiple intervention effects?

Rationale for Scoping Review

A scoping review will map and synthesise the existing evidence base on coenrolment trial designs and statistical methodology. Findings will be organised to create a taxonomy of coenrolment design types and suitable analytical approaches, highlighting their application across different clinical scenarios and outcome data. This will provide the first structured overview of the methodological landscape, identifying gaps and informing subsequent methodological development.

A preliminary search conducted on 12 December 2025 across MEDLINE (PubMed), Scopus, Web of Science, and the Cochrane Database of Systematic Reviews identified no existing or ongoing reviews on this topic, confirming the originality and necessity of this work.

Inclusion Criteria

Concept

- Statistical models and techniques
- Analysis methods
- Multiple, concurrent, or sequential intervention effects
- Interaction effects
- Model performance regarding statistical power, accuracy, precision, error rate

Context

- Statistical methodology
- Mathematical modelling
- Experimental designs and analogous settings that facilitate similar techniques to coenrolment, e.g. factorial designs, partial/incomplete factorial, platform designs, umbrella/basket design, policy analysis
- Variations in terminology for coenrolment, e.g. multiple, overlapping, simultaneous enrolment

Evidence sources

- Statistical methods and epidemiology journal articles, reviews
- Clinical trial study statistical analysis plan

Search Strategy

A full search strategy for one database will be included in the scoping review protocol appendix. Any substantive amendments will be documented in consultation with the librarian team.

The search will be conducted in three stages:

1) initial scoping search in MEDLINE (Ovid) with analysis of titles and abstracts from the retrieved literature as well as their index terms
2) comprehensive search across all recommended databases: MEDLINE (Ovid), EMBASE, Web of Science, Cochrane Library, Scopus. Search terms will include all identified keywords and index terms from the initial search.
3) Reference list screening of included sources, supplemented by author contact where clarification or missing data is required.

Source selection will be carried out following the inclusion criteria of the scoping review protocol. Each search will consist of an initial screening of the title and abstract, followed by another screening of the full text. Results of the search will be managed via EndNote and Rayyan will be used to assist and expedite the screening process.

Examples of data to extract are type of coenrolment, clinical specialty/population, coenrolment design, statistical model and/or test used for analysis, primary outcome type, and covariates used for adjustment (if any). The data extracted for each scoping review question will be summarised by frequency counts across sources after re-categorisation to achieve consistency in terminology, if necessary.

Presentation of Results

The coenrolment designs identified in the search will be presented in a series of tables based on corresponding definitions with any differences between definitions noted and explained. All methods for statistical analysis identified in the review will be summarised by listing all available evidence for each method, the motivation for use (if stated) and the assumptions and methods (e.g. software) required for implementation, together with additional factors associated with coenrolment (e.g. rate of coenrolment, treatment exchangeability and sequence).

Work Package 2—Survey on current views and practices of statisticians on coenrolment in clinical trial research

This work package will deliver a structured survey of trial statisticians to capture current practices, design considerations, and perspectives on co-enrolment in clinical trials. The survey will aim to uncover unreported practices and provide empirical evidence on how co-enrolment is currently understood and implemented, informing the development of updated methodological guidance.

Survey research questions

- What statistical practices are currently used by trial statisticians to analyse clinical trials permitting coenrolment?
- What design considerations are applied when planning trials with coenrolment?

Target Population: The survey will target trial statisticians working within UK that are actively involved in the design, analysis, and statistical oversight of clinical trials.

I will search for participants through international networks, including NIHR Statistics Group, UKCRC Statistics Operations Group, and the International Society for Clinical Biostatics. I will also gather potential participants and register their interest from networking opportunities throughout my fellowship at training courses, annual meetings, and academic conferences.

Methods

Ethical approval will be sought prior to any participation in the survey, as required by the University of Warwick. I will undertake training in survey methodology and will update my research plan accordingly. A key focus of the training will be to learn how to use advanced software tools for web-based surveys, such as Qualtrics and Microsoft Forms.

Survey Development

I will generate survey questionnaire domains through the evidence review from work package 1 and consulting with experts in experimental design and statistical trial methodology.

Survey domains will be limited to 5 in order to minimise respondent fatigue(21). Provisional questionnaire domains include: i) professional background and exposure to coenrolment methodology, ii) practices in statistical analysis, iii) trial design considerations, iv) perspectives on coenrolment and v) recommendations on statistical methodology.

Survey items will be generated for each domain and will follow an iterative process of refinement to maximise clarity, remove redundant questions, and ensure that the survey research questions are thoroughly captured within the questionnaire. I will present draft surveys to my supervisory team, my PPI advisory board, and will consult statisticians and colleagues at the Warwick Clinical Trials unit for feedback.

Testing and Deployment

Pilot testing will be conducted internally with 5-10 trial statisticians from the Warwick Clinical Trials Unit where feedback will be used to check the clarity and operationality of the survey. Preliminary survey data will be used to evaluate the effectiveness of the survey, appraise the design, and gather insights into the scope of the survey. Data will be analysed in R and results will be presented via tables with graphical charts for key results.

Work Package 3—Simulation study of experimental design of coenrolment trials and methods to analysis coenrolment data

I will undertake a comprehensive simulation study to explore how the choice of both design and analysis method, identified in work package 1 & 2, will impact the statistical performance of trial results. The full structure of the simulation study will be formulated via a series of mathematical models and code developed for

implementation in two distinct stages(22) i.e., an investigation on study design followed by an investigation on analysis method.

Data-generating mechanisms

A key part of the design model will be the development of stochastic methods to simulate the recruitment of participants into the trials and data accrual. The simplest models for participant recruitment being a Poisson process with a fixed rate or a time-dependent rate(23). Initially, models will be built around multivariate Normal outcome data using the R package mvtnorm(24), and subsequently expanded to other settings such as time-to-event(25) and binary data. A range of previously described recruitment and data accrual models will be used as a basis for this work(26-28).

Simulation study A, target of analysis: study designs

The first stage will focus on trial design through (i) variations in the number of trial interventions and combinations of treatment arms, (ii) extent of coenrolment (i.e., proportion of co-enrolled patients per domain), (iii) choice of randomisation algorithm (e.g. minimisation), (iv) magnitude and direction of main treatment effects and interactions, (v) simultaneity of treatment application, (vi) temporal patterns of data accrual, (vii) choice of outcome (e.g. binary or time-to-event), (viii) effect of unknown or unobserved confounders and (ix) patterns of missingness and deviations from protocol (e.g. treatment switching and exposure of intervention to control patients).

Once the simulation model is established, this will allow data generation from a putative coenrolment platform design with any of the design characteristics described here. Given the number of design options identified, the number of potential coenrolment designs will be large, so a selection process will be required such that a limited number of design types (consistent with feedback from WP1 and 2) will be used in the second analysis stage of the simulation study.

We anticipate, for practical reasons, that the number of design characteristics will be in the range 20 to 60. As a concrete example of the range of designs tested, Trial design A might be a nominal coenrolment trial design based on one that currently exists, e.g. CoReCCT. Trial design B might differ from design A in parameters relating to the likelihood of coenrolment. Trial designs C and D might differ in the combination and arrangement of sequential/concurrent treatment effects (T_A, T_B, etc.) as well as treatment interactions (T_AT_B, T_BT_C, etc.) of varying magnitudes. This process will be continued until all design parameters relating to coenrolment are sufficiently explored (through trial design E, F, etc.).

Simulation study B, target of analysis: analysis methods (estimand/null hypothesis)

The second stage of the simulation study will explore analytical approaches for the platform coenrolment designs from stage one. The estimand(s) of interest include the treatment effect, interaction effect, and other parameters within the data generating mechanism which will be reviewed and updated by the results of work packages 1 & 2. Variations of the estimands due the selected design characteristics will be investigated such as (i) participant likelihood to coenrol, (ii) presence of

interaction effects, (iii) factors affecting treatment efficacy (e.g., severity of illness, protocol deviations, etc.), and (iv) factors relating to coenrolment policy.

The number of coenrolment scenarios tested will be based on the scoping review (WP1) and the survey of statisticians' practices for managing coenrolment (WP2).

Examples of analytical methods include (i) univariable analysis of covariates with coenrolment(25), (ii) statistical testing across predefined coenrolment subgroups (26) and (iii) multivariate regression models with covariate adjustment for coenrolment(27).

Performance Measures

The performance of the study designs in the first stage will be assessed by sample size, expected sample size, statistical power, and precision. The performance of the statistical models will be assessed by bias, empirical standard error, type I error (false positive) rate, statistical power, sensitivity and specificity in covariate selection of the models, and other features highlighted by the scoping review.

Many simulations (>10,000) will be used to understand the balance of power, precision, bias and error rates for a range of parameter setting and coenrolment scenarios.

Coding and Execution

All coding will be carried out in R using RStudio. Good programming practices and folder management will be adhered to throughout the study and checks built into each process. Each function and analysis will be written in separate scripts. Simulations will be developed in iterations, starting with small scale repetitions and building up to larger frequencies until the desired amount of repetitions is achieved.

Only a small sample of the simulated dataset will be used when testing and developing the simulation programs to minimise bias. Once the full program is complete and functional then the full dataset will be used. The results will be summarised by a tabulation of the performance measures across the coenrolment scenarios (trial designs A, B, C, etc.) and the selection of analytical methods.

All code generated by the study will be made available to others for routine use, for instance as an R package.

Results

The outcome of work package 3 will be a comprehensive assessment and summary of the performance (e.g. precision and bias of treatment effect estimates) of the tested statistical analysis methods across a wide range of coenrolment platform designs. The work will highlight in what circumstances methods work well or badly (or not at all) for the selected performance measures and how the performance is related to decisions made at the design stage (e.g. number and pattern of treatment effects) and resulting from the conduct of the trial (e.g. rates of coenrolment or treatment switching). We anticipate that the performance of the methods will vary widely with the design parameters, so the summary will likely resolve into a more

detailed guidance document that will form the basis of a publication for further dissemination.

Work Package 4 – Practical application using CoReCCT data

I will follow the same methodological approach used in work package 3 but will replace the simulated data by the observed outcome data available from CoReCCT.

Target of analysis: selected models

All available data from the CoReCCT platform study will be combined into a single dataset, including details of randomisations to the four individual domain level trials; Awake Prone (pre-intubation), PROTECT Airways (intubation), RELEASE and UK NAVA (post-intubation). The best performing statistical method(s) from work package 3 that are most suitable for the domains will be applied.

Performance Measures

An analysis of the full CoReCCT study data will provide estimates of marginal treatment effects for each of the four interventions, interactions (between treatment combinations), and error measures—e.g. standard errors and confidence intervals. These will be compared and contrasted between (i) selected methods from work package 3 and (ii) with the main CoReCCT analyses undertaken by the trial teams for each of the four domain level trials.

Results

The primary outcome of work package 4 will be a full and comprehensive analysis of the totality of coenrolment information from the CoReCCT study. This includes a summary of the additional information and knowledge gained based on using the selected coenrolment analysis methods defined in work package 3 and an assessment of the utility of the design choices made for CoReCCT.

****Knowledge mobilisation, dissemination and impact****

The four work packages map directly onto at least four publications in open access academic journals with high impact. These are 1) a scoping review of statistical methodology applicable to coenrolment, 2) survey results on current statistical practices surrounding coenrolment in clinical research, 3) a simulation study of trial designs and statistical models, and 4) an analysis of coenrolment data from CoReCCT.

On completion of my scoping review, I will submit abstracts to academic conferences to present and share my research output. I will also gather potential participants for my survey and register their interest through QR codes and networking opportunities at conferences and annual meetings such as the NIHR Statistics Group, and the UK Clinical Research Collaboration (UKCRC) Statistics Operations Group.

I will also undertake a 12-week research placement at the Australia and New Zealand Intensive Care Society Research Centre (ANZICS-RC). During this time, I will attend Clinical Trial Group meetings to continue to share my research and gather feedback from trialists and statisticians.

The R code developed during the simulation study will be published and may have the opportunity to be developed into packages under the supervision of experienced statisticians at the Warwick CTU. Work package 4 will be a direct application of developed methodology onto real trial data of CoReCCT. I will use my knowledge and expertise gained in the design and analysis of coenrolment methodologies to extend to into collaborations and consult statistical advice. Research groups that have mentioned interest in future collaborations include the ANZICS-RC with their upcoming RECOMMEND platform trial and the PRACTICAL platform study team to further develop their analysis methodology for participants co-enrolled across therapeutic domains. There is also a potential collaboration opportunity with the WHiTE study team to develop the platform concept.

I will work closely with my patient and public involvement (PPI) advisory board to develop and deliver two workshops towards the co-production of lay summaries. I will deploy a range of public engagement techniques to develop plain language descriptions of key terms to explain coenrolment for clinical trial research.

I will continue to disseminate my research with members of the public, including regular updates for those that have previously contributed to the development of this research. I will work closely with the Warwick Institute of Engagement (WIE) and Warwick Medical School (WMS) to prepare and host a talk where my research can be shared and discussed with the local community.

****Project management****

I will be responsible for all project management and research governance matters for the duration of the award. I will meet with my primary academic supervisor every 2 weeks and monthly with my extended supervisory team. All meetings will include a progress/actions updates previous actions, insights/data collected, discussion points, and decisions. The agenda will be sent at least one week in advance and minutes will be recorded and approved at the following meeting. I will also meet with my PPIE advisory board at least twice per year. Meetings will be set according to the availability of the advisory board and the agenda will be sent at least two weeks in advance.

The Warwick Medical School Postgraduate Research (WMS PGR) Team will assess and monitor my attendance and engagement with my studies. I will take several formal and mandatory assessments throughout the PhD, at 9, 21, and 36 months. I will also present reports to an advisory panel made of academic experts in my field. The WMS PGR Team will also assist in providing research support along with the University of Warwick's Doctoral College and Library to ensure I am equipped with all the necessary skills to complete a PhD. The WCTU New Business Committee have committed to providing assistance with budgeting finances, administrative and resourcing issues, and with research support through the WCTU Research Support Service (RSS).

****Intellectual property and commercialisation****

Foreground IP and Research data will be owned and managed by University of Warwick in line with the NIHR requirements.

****Training courses****

I will undertake my PhD in Medical Statistics at the University of Warwick and will have access to the Doctoral College and Library. I will enrol on the Research Development Online (RDO) programme each year and select the seminar/workshop series tailored to the stage of the doctoral research.

In year 1, I will create a foundation in research methods as well as prepare for the research placement through the (RDO) Research Methodologies & Practices Series: review methodology, research designs, fieldwork abroad. This will ensure that I maximise my opportunities for network building, gathering research interest, and developing potential collaborations.

I will enrol onto modules hosted by the Statistics Department to reinforce my statistical training: ST960 Statistics Refresher, ST903 Statistical Methods, ST952 Statistical Practice, ST410 Designed Experiments with Advanced Topics, ST924 Graduate Topics in Probability. I will register to the 'Practical Survey Design and Web-Based Methods' by University of Surrey to guide the survey of work package 2.

I will also take training courses in public engagement and research inclusion for my planned work with communities and people. These are: Effective Community Engagement by The Centre for Ethnic Research (CEHR), and the Patient and Public Involvement and Engagement (PPIE) short course by UCL ICTM. Throughout the first year, I will also undertake PPIE training by the NIHR Birmingham Biomedical Research Centre through their online workshops such as Ethnicity Inclusion, Building and maintaining a PPI group, Evaluating PPI, and the PPI Toolkit.

In year 2, I will enrol onto seminars focusing on maintaining research momentum and additional soft skills for the planned workshops with lay people as co-producers. From the (RDO) PhD Essentials Series I will attend sessions on: time management, 7 steps to PhD completion, productivity & perfectionism, organising and running events, and effective research communication. I will register to short courses and available workshops on experimental design methodology, e.g., 'Using simulation studies to evaluate statistical methods' by UCL ICTM, and 'Adaptive and Bayesian designs in real trials' by the MRC Biostatistics Unit. These will provide me with the technical skills required to undertake work packages 3 and 4.

In year 3, my training will focus on completing my PhD thesis and preparing the next steps of my research career. From the (RDO) Writing Support and Career Development Series, I will attend the following: advanced writing for publication, editing & concise writing, thesis structure & organised writing, designing collaborative projects, and research impact.

Throughout the doctoral award, I will undertake leadership training as opportunities arise, such as the Chief Investigator course by the Warwick Clinical Trials Unit. The Warwick Institute of Engagement (WIE) hosts masterclasses and trainings all year round. I will register to the masterclasses on Equity, Diversity and Inclusion in Public

Engagement, Engagement with Sensitive Topics, Delivering Neuroinclusive Events, and Hands on Engagement.

****Placements****

I have successfully secured a 12-week collaborative research placement at the ANZICS-RC—Australia and New Zealand Intensive Care Society Research Centre—at Monash University.

I will be supervised and hosted by Dr Ary Neto Serpa—a highly accomplished intensivist and Director of Intensive Care Research at Austin Hospital, Australia. He is a committee member of the Invasive Mechanical Ventilation (IMV) domain of the PRACTICAL platform study and involved in the operations for the REMAP-CAP trial, two trials of cutting-edge research that are highly relevant in understanding current practices at the forefront of coenrolment methodology.

From weeks 1 to 2, I will deliver a seminar to the ANZICS-RC and neighbouring departments—Biostatistics Unit, Monash University Clinical Trials Centre (MUCTC), and Health Economics Group—to present and introduce myself and my research.

From weeks 3 to 10, I will be shadowing at the ANZICS-RC and Biostatistics Unit where I will gather insights into the operations of the PRACTICAL platform study and REMAP-CAP. I will also be provided an opportunity to work with Dr Lisa Higgins—an Intensive Care Program Lead and Co-Chief Investigator of the RECOMMEND platform trial that has recently received funding that is expected to incorporate a significant coenrolment policy.

In the final weeks 11 to 12, I will have the opportunity to visit and consult methodology experts at neighbouring centres: The George Institute for Global Health, and the Methods and Implementation Support for Clinical and Health Research Hub (MISCH) at the University of Melbourne.

****Building Collaborations****

My research placement at the ANZICS-RC and attendance at the ANZICS-CTG Annual “Noosa” Meeting will play an important role in the exposure and visibility of my research. I will use the network that I build to plan for and design future collaborative research projects, extending to the PRACTICAL study team (University of Toronto), the Canadian Critical Care Trials Group (CCCTG), and other global experts in coenrolment methodology.

I will keep in touch with those that have registered their interest with an invite to partake in the survey of work package 2, as well as provide opportunities to attend future presentations and updates of my research on coenrolment methodology.

I will also work closely with the CoReCCT—the Confederation of Respiratory Critical Care Trials—collaborators, sharing insights gained from my research and consulting statistical advice in the development of statistical analysis plans. This collaboration

may lead to the addition of new domains onto CoReCCT or the development of another platform-style confederation of clinical trials.

I will also collaborate with the Warwick Institute of Engagement (WIE) in planning public engagement events to share research output to local communities and create a culture of public engagement with research at the Warwick Medical School.

****Conferences****

I will attend the highly regarded and pivotal ANZICS-CTG (Clinical Trials Group) Annual “Noosa” Meeting during my 12-week research placement. I will register and submit an abstract for presentation as an opportunity to share my research and garner interest in the survey of work package 2.

I will register and submit an abstract for presentation at the International Clinical Trials Methodology Conference (ICTMC) in September 2028. At this time, I will be able to present the results of my scoping review (work package 1) and survey (work package 2). I will take the opportunity to consult experts in trial methodology and statistics on the upcoming work package 3, as well as conceive ideas to extend my recently completed work.

I will seek additional funding opportunities where possible to attend the NIHR Statistics Group Annual Conference in order to disseminate research, gather feedback on current work, and stay up-to-date with developments in trial methodology research.

****Other****

I will continue my current responsibilities of mentoring and assisting newly-joined trial statisticians. I will have to opportunity to sit on the Warwick Medical School Statistics Helpdesk to tutor postgraduate students and provide statistical advice for their research projects.

I will also be given a module lecture and workshop to deliver each year on the Health Medical Sciences (BSc) course, giving an overview of core statistical methods and thinking. I will also provide teaching support to computer lab sessions of the Master’s in Public Health (MPH) course.

Dr Ary Serpa Neto
Australian and New Zealand Intensive Care Research Centre
Monash University
553 St Kilda Rd
Melbourne, VIC 3004
Australia

June 2025

Doctoral Fellowship Programme
National Institute for Health Research

Re: Doctoral Fellowship Programme – NIHR Doctoral Fellowship Application by Abraham Contreras

Australian and New Zealand Intensive Care Research Centre (ANZIC-RC), Monash University, is pleased to support Abraham Contreras in his application for the NIHR Doctoral Fellowship Award, and would be happy to act as a host institution for a research placement, should the application be successful.

The proposed research placement would take place over a three-month period starting in February 2027. During this time, Abraham will be embedded within our Research Centre to gain experience and insight into the workings of a methods centre, including exposure to the role of project managers and clinical trial operations.

In addition to observational learning, Abraham plans to conduct a research project investigating the impact of co-enrolment in platform trials, leveraging the expertise and resources available within our team.

We are supportive of Abraham's research on co-enrolment methodologies, and confirm that, if successful, we would welcome his placement with our team and facilitate his engagement with ongoing research activities within our group.

Please let us know if any further information is required.

Yours faithfully,

Dr Ary Serpa Neto
Australian and New Zealand Intensive Care Research Centre
Monash University
553 St Kilda Rd
Melbourne, VIC 3004
Australia
ary.serpaneto@monash.edu

UNIVERSITY OF WARWICK

Warwick Clinical Trials Unit
Warwick Medical School
University of Warwick
Coventry
CV4 7AL, UK

9th January 2026

Dear NIHR Doctoral Awards Committee

RE: Abraham Contreras

The University of Warwick, Warwick Medical School (WMS) and Warwick Clinical Trials Unit (WCTU) fully support Abraham Contreras's NIHR Doctoral Award. This statement has been prepared by Professor Nigel Stallard, Director of WCTU.

WCTU and the University of Warwick is an excellent environment for Abraham to successfully complete this doctoral fellowship. Warwick CTU has extensive expertise in research and clinical trials of complex interventions in emergency and critical care medicine and surgery, pain and rehabilitation and statistical and methodological research in the design, conduct and analysis of data from clinical trials. Abraham's proposal to understand the current application and to develop robust, modern statistical methodology and guidance to support the design and analysis of clinical trials of complex interventions that incorporate co-enrolment, fits well with our trials portfolio and with the research interests of colleagues at Warwick CTU. The CTU has been one of the most successful in the UK by NIHR funding over recent years and emergency and critical care medicine and surgery, pain and rehabilitation trials are a major contributor to that success.

To directly support Abraham's research, we will provide access to high-performance computing resources, licensed statistical software, and secure data environments for simulation and analysis. He will also be provided with structured administrative support throughout the fellowship, including assistance with public engagement set-up, ethics, data governance, and quality assurance. Dedicated staff from the Warwick Medical School PGR team will conduct regular formal reviews with external experts to provide thorough appraisal of research quality and ensure timely progress against milestones.

Abraham has been an employee at the University of Warwick since July 2021. He is based at WCTU, within Warwick Medical School, where he a Research Associate within the Statistics group. Abraham is a bright and ambitious researcher who has shown clear potential to develop into a distinguished research leader in the future. He demonstrates a strong intellectual curiosity and excellent research and analytical skills, the ability to collaborate with researchers and a commitment to develop his own agenda of impactful research in the future. Abraham will be part of a large and highly successful research group, based at

the University of Warwick, with extensive research expertise in both trials methodology, statistical methods and clinical trial design and practice.

WCTU consists of a strong support network consisting of academic AHPs, surgeons, trial methodologists, statisticians, health economists and other research professionals with a strong track record in world-leading research aiming to improve the health and wellbeing of the population. Working within this team will help Abraham develop a wide experience in the process of developing, implementing and analysing data from a clinical study. To nurture Abraham's long-term research career, WCTU will implement a formal mentorship structure comprising a personal mentor, Dr Chen Ji, and an external clinical collaborator, Prof Louise Rose. He will be encouraged to take on leadership roles as they arise within the unit to enhance his visibility and networks.

WCTU is an accredited professional organisation, which prospers on the development of its staff. The organisation is made up of over 100 research staff and within the Statistics Group there are 2 professorial staff, 3 associate professors, 1 senior research fellow, 2 research fellows and 4 associate research fellows. The Unit is home to a large number of researchers working closely with NHS stakeholders, patients and public to assist in medical research, which depends in large part on the use and development of new statistical methods.

As part of Abraham's training package, he will collaborate within WMS with Warwick Applied Health, and with the University Statistics Department. In summary, WCTU, WMS and the University of Warwick offer excellent training and development opportunities, which are a main motivation and objective of Abraham's Fellowship. Abraham will have access to a range of important resources including: (a) an assortment of historical and contemporary clinical studies/data, including complex intervention trials, which used co-enrolment directly or can be used as the basis for simulating co-enrolment; (b) access to leading academic staff members through, coaching and mentoring schemes, to assisting in developing ideas and collaborating on work packages within the PhD project; (c) courses and workshops aimed specifically at PhD students (e.g. study skills and academic writing) and University programs on leadership and development; (d) frequent Unit-based, Department and University-wide seminars and meeting (reading groups), with opportunities for Abraham to present his work to interdisciplinary audiences. This will offer Abraham valuable support to carry out novel research in the future and is in clear and close alignment with the Unit and Abraham's trajectory of development and his overarching aim to develop guidelines and establish appropriate methodology for the use of co-enrolment in clinical trials.

Abraham's supervisory team at WCTU have all supervised PhD candidates to completion and have experience of supervising successful NIHR doctoral fellows.

Professor Nick Parsons has extensive experience in both the conduct and reporting of clinical trials and also the assessment and the development of novel statistical methods, with a particular focus on the use of simulation and software implementation. Professor Ranjit Lall will provide statistical insights into the application of co-enrolment in clinical trials and also their application and use in emergency and critical care medicine, where she has extensive experience. Particular related to the ongoing clinical trials platform Confederation of Respiratory Critical Care Trials (CoReCCT; Awake Prone ISRCTN 63784375, Protect Airways ISRCTN 15882946, Release ISRCTN 17158033, NAVA) which uses co-enrolment and where she is the statistical lead. Dr Chen Ji is a senior trials statistician and is lead for CoReCCT and as such will provide direct input to the application of co-enrolment in this study.

The University of Warwick strives towards creating an equal and safe working environment for people from all walks of life. Warwick works towards several charters to measure and benchmark our progress on equality, diversity and inclusion including Athena Swan, Disability Smart, HR Excellence in Research Award, and Stonewall UK Workplace Equality Index. In committing to the principles of the Athena Swan Charter, and holding a Silver Award status, The University of Warwick recognises that we join a global community with a shared goal of addressing gender inequalities and embedding inclusive cultures. Warwick ranked 123 out of 500 in the 2020 Stonewall index. Furthermore, Warwick has developed safeguarding policies in accordance with the Concordat to Support Research Integrity and that align with NIHR values. I believe Abraham will thrive in this inclusive and diverse environment.

Overall, I am pleased to support Abraham in this application and can confirm that Warwick CTU will provide the ideal settings for him, with the expertise, environment and administrative support for him to complete this fellowship successfully and to advance his career as an academic medical statistician in the future.

Nigel Stallard.

Professor Nigel Stallard
Director of Warwick Clinical Trials Unit & Professor of Medical Statistics

NIHR Doctoral Award - Gantt Chart

Lead Applicant: Abraham Contreras

REFERENCES

1. Krige A, Pattison N, Booth M, Walsh T, Walsh T, Fletcher S, et al. Coenrolment to Intensive Care Studies – A UK Perspective. Journal of the Intensive Care Society. 2013;14(2):103-6. doi: 10.1177/175114371301400203.
2. Felton T, Pattison N, Fletcher S, Finney S, Walsh T, Dark P. Coenrolment to UK Critical Care Studies - A 2019 update. J Intensive Care Soc. 2022;23(1):53-7. doi: 10.1177/1751143720971542.
3. Clarke F, Hand L, Deane A, Zytaruk N, Hardie M, Arabi Y, et al. Coenrollment in a critical care trial: Characteristics and consequences. Contemporary Clinical Trials. 2025:107938. doi: https://doi.org/10.1016/j.cct.2025.107938.
4. Practical PTI, investigators R-C. The Rise of Adaptive Platform Trials in Critical Care. Am J Respir Crit Care Med. 2024;209(5):491-6. doi: 10.1164/rccm.202401-0101CP.
5. Angus DC, Mira JP, Vincent JL. Improving clinical trials in the critically ill. Crit Care Med. 2010;38(2):527-32. doi: 10.1097/CCM.0b013e3181c0259d.
6. Larntz K, Neaton JD, Wentworth DN, Yurik T. Data analysis issues for protocols with overlapping enrollment. Stat Med. 1996;15(21-22):2445-53; discussion 55-8. doi: 10.1002/(sici)1097-0258(19961130)15:22<2445::aid-sim462>3.0.co;2-8.
7. Nichol G, Powell JL, Emerson S. On coenrollment in clinical resuscitation studies: review and experience from randomized trials. Resuscitation. 2010;81(7):792-5. doi: 10.1016/j.resuscitation.2010.03.014.
8. Tetzlaff JM, Chan AW, Kitchen J, Sampson M, Tricco AC, Moher D. Guidelines for randomized clinical trial protocol content: a systematic review. Syst Rev. 2012;1:43. doi: 10.1186/2046-4053-1-43.
9. Reade MC, Bass F, Howe B, Seppelt I, Shehabi Y. Considerations for coenrolment in randomised controlled effectiveness trials in critical care: the SPICE-8 coenrolment guidelines. Crit Care Resusc. 2017;19(2):110-4.
10. Maslekar T, Peat S, Karim J, Moriarty C, Jayne DG, Chapman SJ. The reporting of coenrolment in protocols of publicly funded randomized controlled trials was infrequent and variable. J Clin Epidemiol. 2021;133:94-100. doi: 10.1016/j.jclinepi.2021.01.002.
11. Burnet K, Benson J, Earl H, Thornton H, Cox K, Purushotham AD. A survey of breast cancer patients' views on entry into several clinical studies. Eur J Cancer Care (Engl). 2004;13(1):32-5. doi: 10.1111/j.1365-2354.2003.00439.x.
12. Morley CJ, Lau R, Davis PG, Morse C. What do parents think about enrolling their premature babies in several research studies? Arch Dis Child Fetal Neonatal Ed. 2005;90(3):F225-8. doi: 10.1136/adc.2004.061986.
13. Cook DJ, Ferguson ND, Hand L, Austin P, Zhou Q, Adhikari NK, et al. Coenrollment in a randomized trial of high-frequency oscillation: prevalence, patterns, predictors, and outcomes*. Crit Care Med. 2015;43(2):328-38. doi: 10.1097/CCM.0000000000000692.
14. Costa ML, Achten J, Appelbe D, Athwal A, Grant R, Cook J, et al. The World Hip Trauma Evaluation (WHiTE) platform trial: a framework for randomized comparisons of interventions for fragility hip fracture. Bone Jt Open. 2025;6(4):383-90. doi: 10.1302/2633-1462.64.BJO-2024-0240.
15. Cafferty FH, Coyle C, Rowley S, Berkman L, MacKensie M, Langley RE. Coenrolment of Participants into Multiple Cancer Trials: Benefits and Challenges. Clin Oncol (R Coll Radiol). 2017;29(7):e126-e33. doi: 10.1016/j.clon.2017.02.014.

16. Rose L, Camporota L, Contreras A, Couper K, Hadfield D, Alleyne S, et al. The COnfederation of REspiratory Critical Care Trials (CoReCCT): a UK research confederation. Br J Anaesth. 2025;134(2):566-8. doi: 10.1016/j.bja.2024.10.033.
17. Heath A, Yarnell C, Xu W, Urner M, Goligher E, Fan E, et al. Master Statistical Analysis Plan for the PRACTICAL Platform2022 3rd June 2025. Available from: https://practicalplatform.org/docs.
18. Cook DJ, Blythe D, Rischbieth A, Hebert PC, Zytaruk N, Menon K, et al. Enrollment of intensive care unit patients into clinical studies: a trinational survey of researchers' experiences, beliefs, and practices. Crit Care Med. 2008;36(7):2100-5. doi: 10.1097/CCM.0b013e31817c00b0.
19. Peters MDJ GC, McInerney P, Munn Z, Tricco AC, Khalil, H Scoping Reviews (2020). 2024 26 March 2025 [cited 15 May 2025]. In: JBI Manual for Evifence Synthesis [Internet]. JBI: JBI, [cited 15 May 2025]. Available from: https://synthesismanual.jbi.global/.
20. Open Science Framework [Available from: https://osf.io/.
21. Burns KE, Duffett M, Kho ME, Meade MO, Adhikari NK, Sinuff T, et al. A guide for the design and conduct of self-administered surveys of clinicians. CMAJ. 2008;179(3):245-52. doi: 10.1503/cmaj.080372.
22. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. Stat Med. 2019;38(11):2074-102. doi: 10.1002/sim.8086.
23. Parsons NR, Basu J, Stallard N. Group sequential designs for pragmatic clinical trials with early outcomes: methods and guidance for planning and implementation. BMC Med Res Methodol. 2024;24(1):42. doi: 10.1186/s12874-024-02174-w.
24. mvtnorm: Multivariate Normal and t Distributions [Internet]. 2025. Available from: https://cran.r-project.org/package=mvtnorm.
25. reda: Recurrent Event Data Analysis [Internet]. 2022. Available from: https://github.com/wenjie2wang/reda.
26. Barnard KD, Dent L, Cook A. A systematic review of models to predict recruitment to multicentre clinical trials. BMC Med Res Methodol. 2010;10:63. doi: 10.1186/1471-2288-10-63.
27. Gkioni E, Rius R, Dodd S, Gamble C. A systematic review describes models for recruitment prediction at the design stage of a clinical trial. J Clin Epidemiol. 2019;115:141-9. doi: 10.1016/j.jclinepi.2019.07.002.
28. Gkioni E, Dodd S, Rius R, Gamble C. Statistical models to predict recruitment in clinical trials were rarely used by statisticians in UK and European networks. J Clin Epidemiol. 2020;124:58-68. doi: 10.1016/j.jclinepi.2020.03.012.